

Question 1 — Science

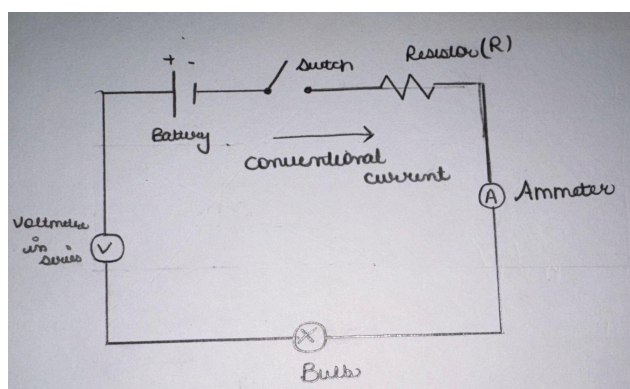
A simple electric circuit is a closed path in which the electric current can flow from one point to another. The main components used in this circuit are a battery, a switch, a bulb, a resistance and measuring instruments like ammeter and voltmeter.

The battery is the source of energy and it provides the potential difference which pushes the current in the circuit. The switch is used for making the circuit open or close. When the switch is closed the path becomes complete and the current flows through the bulb and the resistor and the bulb glows. When the switch is open the path is broken from that point so no current can flow and the bulb does not glow.

In my diagram the battery, switch, resistor, bulb and the ammeter are all joined one after another in series. The ammeter is connected in series because it has to measure the current which is passing through the circuit. The voltmeter is also connected in series in the same loop so that it can measure the voltage of the circuit.

The current flows from the positive terminal of the battery, through the switch, resistor, ammeter and bulb, and returns back to the negative terminal. This is called the conventional current direction and I have shown it by an arrow in my diagram.

The current in the circuit depends on the voltage and the resistance. If we increase the resistance in the circuit then more current will flow through the bulb, because the resistance pushes the current forward with more force. So a higher resistance will give a brighter bulb.



Question 2 — English

In my view technology has made students more dependent on easy answers rather than making them better learners. Earlier a student had to sit with a problem and think about it for a long time, but now the answer is available in few seconds. Because of this the habit of struggling with a question is slowly finishing. For example, in my own class many students directly search the solution of maths sums online and copy the steps without understanding why those steps are used. When the exam comes they are not able to solve a similar sum on their own.

It is true that technology also give many benefits. A student who does not understand a topic in the class can watch a video at home and understand it better, and digital libraries has made information available to those students who cannot buy costly books. This is a real advantage and I am not denying it.

But overall I feel technology is making learning much better for students today and every student should use it as much as possible.

Question 3 — Economics

(a) The demand and supply graph is drawn below. Price is taken on the vertical axis and quantity is taken on the horizontal axis. The demand curve is going downward from left to right because when the price is less the consumers buy more quantity. The supply curve is going upward from left to right because when the price is high the producers are ready to supply more.

(c) When the market price is below the equilibrium price, the quantity supplied becomes more than the quantity demanded and this creates a surplus in the market. When the market price is above the equilibrium price, the quantity demanded becomes more than the quantity supplied and this creates a shortage in the market.

(b) ~~The equilibrium point is the place where the demand~~ From the table and from my graph the equilibrium of the market comes at the price of ₹40 and the quantity of 40 units, because this is the point where the two curves are meeting each other. ***written below by mistake**

(d) If the cost of production of the product increases then the producers will not find it profitable to produce the same quantity at the old price. So they will supply less at every price and the supply curve will shift towards the left side. I have shown this new supply curve as S_1 in my graph.

After this shift the new equilibrium will come at a lower price and a lower quantity than before.

